

CORE COMPETENCIES

- Python, PyTorch, SciKit-Learn ML
- LLM Prompt Engineering, RAG Tuning
- AI App. Dev.
- Western Blot, PCR, Chromatin Assay
- Yeast Genotyping & Transformation
- Light Spectroscopy
- Dropout & Liquid Media Preparation
- Competent Cell Preparation

EDUCATION:

University of California: San Diego
(Bioengineering: Bioinformatics B.S.)

Fall 2024 - Present
Expected Graduation: Spring 2028

EMPLOYMENT & RESEARCH:

Zhou Lab (UCSD Cellular Molecular Medicine)
(Lab Assistant)

Fall 2024 - Fall 2025

- Supported graduate-level research by conducting molecular biology techniques such as spotting and chromatin assays, protein purifications, SnapGene primer design, and tetrad dissections.
- Performed general maintenance tasks, including autoclave, yeast culturing, preparing dropout and liquid media, and competent cell preparation. Followed compliance and safety protocols..
- Collected and analyzed data, presented journal clubs during lab-wide meetings

Polygence Research Project
(Researcher)

Spring 2023 - Fall 2023

- Used ML algorithms with Python's PyTorch and Scikit libraries to analyze oncological datasets. Applied image classification analysis to examine histopathologies & tumor CT scans.
- Wrote predictive algorithms for detecting breast tumor malignancy and invasive ductal carcinomas. Presented research at Polygence Symposium.

INTERNSHIPS & PROJECTS:

Amazon Web Services AI Summer Camp collab. w/ Cal Poly DxHub
(Intern)

Summer 2025

- Selected as 1 of 80 students from over 1,300 applicants in the California CC cohort.
- Developed an AI chatbot using AWS Cloud Services to help students explore career paths and majors available at their Community College; Integrated Holland Code into chatbot's logic.
- Gathered feedback on prototype and UI from Santa Rosa CC Executive Dean.
- Delivered chatbot to AWS & Cal Poly DxHub panel; Project earned 3rd place out of 16 teams.

BioMedical Engineering Society @ UCSD
(Bioinformatics Workshop Lead - BioWorks Committee)

Fall 2025 - Present

- Leading bioinformatics workshops on R data visualization, PyTorch model implementation by collaborating with committee co-chairs on design and execution.
- Developing hands-on projects using public Kaggle genomic datasets to demonstrate workflows in sequencing analysis, Alpha Fold, and machine learning-based data exploration.

NanoEngineering & Technology Society (NETS) @ UCSD
(Synthetic NanoSkin Project Member)

Fall 2025 - Present

- Project Overview: Design and test a cost-effective Synthetic Nanoskin Bandage. Collaborating with NanEngineering Faculty.
- Conducting Tissue Engineering literature review to develop a biocompatible chitosan-gelatin crosslink protocol for wet-lab prototyping and experimentation in the Winter Quarter.

CardinalBotics (FIRST Robotics Competition 4159)
(Head of Training and Member Onboarding, Head of Competition Scouting)

Fall 2021 - Spring 2024

- Managed the administration of the team's Internal Robotics Competition by leading 18 training workshops, teaching 60 new members robot programming basics and power tool safety.
- Organized and led 35+ robot demonstrations and cellular biology workshops across the Bay Area with LEGO EV3 Kits and Foldscope Instruments; Reached 15,000+ new beneficiaries.